

NISHIL PATEL

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Professional Summary

AI Engineer with hands-on experience shipping AI-powered applications. Built enterprise GenAI services with retrieval that returns the right chunk and backends that stay reliable under real usage. Focused on engineering production-ready AI systems with optimized LLM pipelines, latency optimization and robust failure handling.

Technical Skills

- Programming:** Python, SQL
- Generative AI:** Large Language Models (LLMs), Prompt Engineering, Context Engineering, AI Agents, Retrieval-Augmented Generation (RAG)
- AI Frameworks:** Vanna AI, LangChain, LangGraph, CrewAI, Flowise AI, Model Context Protocol (MCP)
- LLMOps:** Prompt Versioning, Token/Cost Optimization, LLM Evaluation, Caching Strategies, Latency Optimization
- LLM Platforms:** Anthropic Claude, OpenAI GPT, Google Gemini, xAI Grok, Moonshot AI Kimi, Zhipu AI GLM
- Backend & Databases:** FastAPI, Flask, REST APIs, Pydantic, Vector Databases, PostgreSQL, MySQL, NoSQL
- Cloud & DevOps:** Microsoft Azure, Docker, Apache Airflow, Git, GitHub

Professional Experience

FortifAI Inc. July 2025 – Present
AI Engineer Ahmedabad, India

- Built and maintained production backend for **D7**, an AI-powered due diligence product that orchestrates Anthropic Claude and OpenAI models to generate structured enterprise risk reports.
- Developed D7's Quality Control (QC) automation as configurable workflows for due diligence report validation.
- Designed a structural compliance platform using Langchain for document retrieval and partnered with the Data Engineering Team on Apache Airflow orchestration supporting enterprise AI data pipelines.

Alchemyte Data Solutions January 2025 – June 2025
Data Science Engineer Intern Ahmedabad, India

- Built an LLM-powered fraud detection assistant using Python, SQL and Vanna AI owning the prompt engineering layer with GPT-4 for natural language-to-SQL generation over PostgreSQL-backed procurement schemas, improved query reliability and accuracy for audit and compliance teams.

Projects

D7 - AI Powered 3rd Party Due Diligence Python, FastAPI, PostgreSQL, Claude, OpenAI

- Developed the backend of a production AI-powered due diligence platform that transforms structured analytical data into dimension-specific narratives, risk scores and enterprise risk reports using LLMs.
- Optimized LLM inference through intelligent JSON chunking, context engineering, prompt optimization and API response caching to reduce token consumption while preserving output quality.
- Engineered an automated QC framework to audit generated reports in real time, ensuring all narratives, numerical values and risk scores were accurately grounded in the source data.

Enterprise AI Compliance Assistant Python, FastAPI, OpenAI, LangChain, PostgreSQL

- Architected a compliance platform using LangChain and evaluated LangGraph to orchestrate enterprise document retrieval, risk analysis and structured compliance report generation.

Publications

Forecasting Maximum Temperature Trends with SARIMAX: A Case Study from Ahmedabad, India MDPI
Cipherring the Modern World: A Comprehensive Analysis of DES, AES, RSA, and DHKE IEEE
Mastering Data Analysis and Data Visualization with Time Series Techniques in Python MEDIUM

Education

Pandit Deendayal Energy University (PDEU) Gandhinagar, India 2021 – 2025
B.Tech. in Computer Engineering CGPA: 8.67/10